

Episcleral plaque radiotherapy for uveal melanoma

Michael E. Giblin FRACO, FRACS

Oncology Service, Wills Eye Hospital, Jefferson Medical College, Thomas Jefferson University, Philadelphia

Jerry A. Shields MD

Oncology Service, Wills Eye Hospital, Jefferson Medical College, Thomas Jefferson University, Philadelphia

James J. Augsburger MD

Oncology Service, Wills Eye Hospital, Jefferson Medical College, Thomas Jefferson University, Philadelphia

Luther W. Brady MD

Department of Radiation Oncology and Nuclear Medicine, Hahnemann University, Philadelphia

Abstract

The use of radioactive Iodine, Ruthenium, Iridium and Cobalt plaques of varying dimensions to treat choroidal and ciliochoroidal melanomas is discussed and placed in perspective with other treatment modalities. The regression pattern following plaque treatment appears to be more rapid for tumours that are more poorly differentiated. The visual results following plaque brachytherapy compare very favourably with those following local resection and external beam radiation. Ocular complications are significantly greater for tumours whose initial thickness exceeds 8 mm. The survival rate after plaque treatment is compared with that for patients who have had primary enucleation for equivalent-sized tumours.

Key words: Choroidal melanoma, ciliochoroidal melanoma, plaque radiotherapy, brachytherapy, regression, survival.

Primary enucleation has been the traditional method of management of patients with choroidal and ciliochoroidal melanomas; however, the effectiveness of enucleation for preventing metastases has been questioned since evidence presented by Zimmerman and others indicated that enucleation may in fact accelerate extraocular dissemination of tumour cells.^{1,2} Consequently, other techniques to control tumour growth have gained in popularity. When properly employed, these techniques can preserve the eye and permit

useful vision. Included amongst these is radioactive episcleral plaque therapy.

As early as 1930, Foster Moore³ described the intraocular insertion of Radon seeds to treat a uveal melanoma. In 1966, Stallard reported his results using high-energy radioactive Cobalt plaques.⁴ Subsequently Lommatzsch popularised the use of low-energy Ruthenium plaques,⁵ and more recently low-energy Iodine⁶ and Iridium plaques have been introduced.⁷ These newer, low-energy plaques are intended to produce fewer radiation-induced ocular complications; however, Cobalt 60 remains a useful radioisotope for the treatment of tumours of larger thickness.⁷

Pigmented choroidal and ciliochoroidal lesions may be arbitrarily labelled for clinical purposes as naevi if less than 2 mm thick, and melanomas if greater. It is the thickness of the tumour and not the maximal basal dimension which is the critical factor in this respect.⁸

For tumours less than 2 mm thick and those greater than 2 mm thick but exhibiting dormant characteristics including overlying drusen and fibrous metaplasia of the retinal pigment epithelium, regular observation is now widely accepted as the most appropriate form of management.⁸⁻¹⁰ Laser photocoagulation has been used to treat small tumours, but is now seldom indicated because many of those lesions that would have been treated with this modality show no signs of growth with periodic observation.^{9,11} Where tumours are greater than 2 mm thick without dormant features and for those where unequivocal growth has been

documented, definitive treatment is generally indicated.

For tumours less than 10 mm maximal basal dimension and lying anterior to the equator, local resection may be considered.^{9,12} Some ophthalmic surgeons resect larger and more posterior tumours; however, the surgery is more difficult and visual results no better than with plaque brachytherapy in the majority of such cases.¹³

Proton beam and helium ion radiation are acceptable alternatives to radioactive plaque treatment for posterior uveal melanomas. These techniques offer the theoretical advantages of highly localised, fractionated radiation with uniform tumour radiation distribution; however, the visual results do not appear to be significantly better and anterior complications including radiation keratitis, radiation dermatitis, loss of cilia and punctal stenosis are greater following these modalities than with plaque brachytherapy. These treatment forms are relatively expensive, do not avoid surgery and seem to be limited by the ability of the patient to maintain steady fixation during multiple treatment sessions.¹⁴⁻¹⁶

For those tumours less than 20 mm maximal basal dimension and considered unsuitable for local resection, radioactive plaque treatment may be considered.⁷ Anterior chamber extension, retinal invasion, rupture of Bruch's membrane, peripapillary location and extraocular extension are not necessarily contraindications to plaque brachytherapy.

Enucleation is usually advised for those patients with large melanomas whose maximal basal dimension exceeds 20 mm, or for smaller tumours where there is clinical evidence to suggest optic nerve invasion. Plaque brachytherapy has occasionally been used to control tumour growth in cases where the maximal basal tumour dimension exceeded 20 mm; however, the subsequent incidence of radiation-induced ocular complications is unacceptably high in most of these patients.

Four plaque types are used on the Oncology Service at Wills Eye Hospital, namely Iodine 125, Iridium 192, Ruthenium 106 and Cobalt 60. All are gamma ray emitters except for Ruthenium 106 which is primarily a beta ray emitter.

In general, Iodine 125 plaques have been used for melanomas 2 to 3 mm thick, Iridium 192 or Ruthenium 106 for tumours 3 to 5 mm thick and Cobalt 60 for tumours greater than 5 mm thick. This selection is based upon the so-called depth-dose radiation distribution characteristics of the different radioisotopes corresponding to the geometry of the particular plaque. In general, the

apex of the irradiated melanoma receives 8000 to 10 000 cGy (rads) and the base not more than 40 000 cGy. The choice between Ruthenium 106 and Iridium 192 depends upon the location of the tumour. Ruthenium 106 is preferred for ciliochoroidal melanomas because it is shielded and because the shorter wavelength beta emission causes less damage to the lids, cilia, cornea and lens than the longer wavelength gamma radiation of Iridium 192. For choroidal melanomas 3 to 5 mm thick, either Iridium 192 or Ruthenium 106 plaques are used.

Iodine 125 and Iridium 192 plaques may be fashioned to different shapes for special situations. Iodine 125, Iridium 192 and Ruthenium 106 are also available as notched plaques to accommodate tumours situated less than 2 mm from the optic nerve head.

Although 15 mm is the most commonly used plaque diameter, 10 mm, 20 mm and 25 mm plaques are available depending upon the individual isotope used. A 15 mm diameter plaque with a posterior 10 mm active isotope distribution is sometimes of use for small, posterior tumours threatening central vision. This type of plaque offers highly localised radiation, and is considerably easier to suture to the globe than a standard 10 mm plaque.⁷

As part of the preoperative evaluation indirect ophthalmoscopy is performed and a fundus diagram is made. Anteriorly situated, pigmented melanomas are transilluminated in order to confirm the tumour's maximal basal dimension, which dictates whether local resection or plaque brachytherapy is more appropriate. Gonioscopy is performed for ciliochoroidal melanomas to determine if anterior chamber extension is present. Fundus photographs and ultrasonographic thickness measurements serve as essential baseline studies, the former to delineate clearly the border relationships of the melanoma and the latter to establish the type of radioisotope to be used and the duration of plaque application. Fluorescein angiography is arguably the least contributory of the ancillary studies performed.

At surgery, the tumour is localised in most cases using either transscleral or transcorneal transillumination. Amelanotic melanomas are localised using indirect ophthalmoscopy and scleral depression. A dummy plaque is then applied to facilitate accurate suture placement. The live plaque is then secured to the globe and left in place until the appropriate radiation dose has been delivered according to computer-generated calculations based on the individual radioisotope being used and the tumour thickness.

Post-treatment ultrasound thickness measurements are performed every four to six months and when compared with the initial ultrasound measurement are helpful in determining regression, which usually involves a diminution in tumour thickness rather than diminution in basal dimensions. Regression patterns following plaque treatment fall into four groups:

1. A slow decrease in tumour thickness as determined by serial ultrasonographic measurements.
2. Rapid shrinkage with almost total disappearance of the tumour over one to three years.
3. Shrinkage with exudative response, especially in eyes with pretreatment bullous retinal detachments.
4. Shrinkage with haemorrhagic response.^{9,17}

Augsburger, Shields *et al*¹⁸⁻²⁰ reported that melanomas initially managed by a period of observation, which were documented to grow rapidly prior to enucleation, were more likely to contain epithelioid cells than those which grew slowly. Based on these findings, it is assumed that tumours of equivalent size which demonstrate rapid growth rates prior to radioactive plaque application also contain predominantly epithelioid cells. These same rapidly growing tumours have been found to display a correspondingly rapid regression pattern following plaque radiotherapy. For this reason, rapid shrinkage of a melanoma following plaque brachytherapy is considered a less favourable sign in terms of systemic prognosis than slow shrinkage.²¹

Although post-treatment visual acuity appears to match that following proton beam or helium ion radiation, and local resection, there are no reported large series of patients managed by these various methods in which visual results are directly comparable. The visual outcome depends upon the location of the tumour and its size, which are indirect determinants of the foveal and disc doses of radiation, and the presence and duration of foveal retinal detachment. The visual prognosis appears to be relatively good if the foveal radiation dose is less than 5000 cGy. In a review of 77 patients treated with Cobalt 60 plaque brachytherapy, 70% retained visual acuity better than 20/200 three and a half years after treatment and 62% had visual acuity better than 20/50.¹⁴

Ocular complications include radiation vasculopathy of the directly radiated field, radiation retinopathy and papillopathy, vitreous haemorrhage, cataract, neovascular glaucoma, radiation keratopathy, dry eye, radiation blepharitis and scleral necrosis. When compared with results following external beam radiation, posterior

segment complications seem to occur with a marginally greater frequency following plaque radiotherapy, while anterior segment and external ocular complications appear to be greater following external beam radiotherapy.^{8,9,16}

Statistical evidence available from the Oncology Service at Wills Eye Hospital indicates that 60% of eyes in which the initial tumour thickness is 8 mm are retained five years after plaque brachytherapy. This figure drops to 30% ocular retention at five years for tumours with an initial thickness of 12 mm. The reason for enucleation in these cases is usually painful glaucoma and not tumour regrowth.²²

Survival rates following plaque radiotherapy are now emerging.^{23,24} So far these show no appreciable difference in ten-year survival between those patients managed with Cobalt 60 plaque brachytherapy and those managed by enucleation.²⁴

Ultrasonic hyperthermia is presently being used in conjunction with radioactive plaque therapy in a small number of centres. The rationale for this treatment is that less radiation is necessary to control the growth of tumour cells whose temperature is elevated using ultrasound. It is hoped that this technique will result in fewer radiation-induced ocular complications and therefore further improvement in long-term visual acuity results; however, the practical value of this form of treatment has not yet been established.²⁵⁻²⁸

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